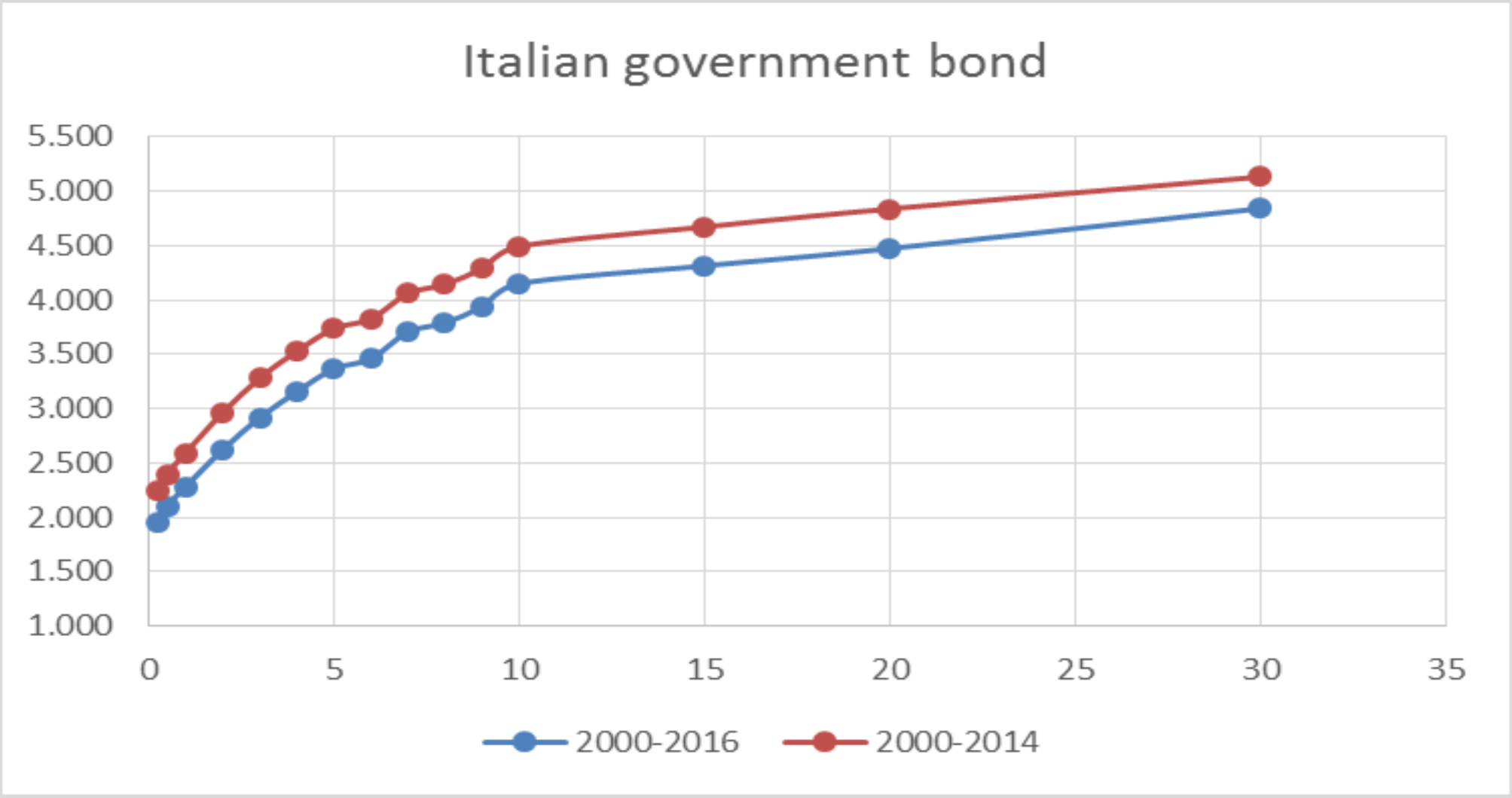
# THE TERM STRUCTURE OF INTEREST RATES

FINANCIAL ECONOMICS – SPRING 2026

# DEFINITIONS

- In the market, there is not a single interest rate, but an interest rate for each time maturity. The spot rate (or zero rate) for a maturity T is the interest rate associated with the price of a security at a discount for that maturity, with the symbol $i_T$
- The 'short-term rate' is the prevailing rate on bonds with an annual maturity and the 'long-term rate' is the prevailing rate on ten-year bonds

- The "term structure" is the whole set of rates prevailing over the various maturities. The typical shape is concave, with long rates higher than short rates


Italian government bond
5.500
5.000
4.500
4.000
3.500
3.000
2.500
2.000
1.500
1.000
0
5
10
15
20
25
30
35
2000-2016
2000-2014

# PRICE OF BOND WITH A TERM STRUCTURE

- for a ZCB the price is the present discounted face value:

$$P_T = \frac{FV}{(1 + i_T)^T}$$

- A bond that pays coupons can be interpreted as a portfolio of zero-coupon bonds. For example, a two-year bond with a coupon $C$ and face value $FV$ is a portfolio composed of a ZCB with face value $C$ and a ZCB with face value $C + FV$. Therefore

$$P = \frac{C}{(1 + i_1)} + \frac{C + FV}{(1 + i_2)^2}$$

# EXAMPLE

- Compute the value of a bond with a coupon rate of 10% and a nominal value of 1,000, using the following spot rate structure: 5% at 1 year, 6% at 2 years, 7% at 3 years

$$P = \frac{100}{1.05} + \frac{100}{(1.06)^2} + \frac{1{,}100}{(1.07)^3} = 1{,}082.17$$

# EXAMPLE

- The YTM y is:

$$1{,}082.17 = \frac{100}{(1+y)} + \frac{100}{(1+y)^2} + \frac{1{,}100}{(1+y)^3}$$

To get $= 6.88\%$. The YTM is different from the interest rate used for the third-year cash flow discount

# COMPUTING THE TERM STRUCTURE

- The function representing the term structure is calculated using primarily information from the ZCBs
- The information provided by ZCBs can be used as part of a process called "bootstrapping", which uses both ZCBs and coupon securities
- Example: the 1-year spot rate is 5% and the two-year spot rate is 6%, as in the example above, and we know that the price of a bond with a three-year coupon is 1,082.17
- How do we infer the three-year spot rate? By definition it must be that rate $i_3$ such that:

$$1{,}082.17 = \frac{100}{1.05} + \frac{100}{(1.06)^2} + \frac{1{,}100}{(1+i_3)^3}$$

Solving we find 7%

# FORWARD RATES

- The forward rate is a rate that is applied to a future maturity. Forward rates can be calculated from spot rates using the following definition

$$(1 + f_n) \quad = \frac{(1 + i_n)^n}{(1 + i_{n-1})^{n-1}}$$

- Rewriting as $(1 + i_{n-1})^{n-1}(1 + f_n) \quad = (1 + i_n)^n$

- The forward rate can therefore be interpreted as the prevailing remuneration in a future period n with the following property: investing a sum of money for n-1 periods at the rate $i_{n-1}$ and for one period at rate $f_n$ produces an amount identical to that produced by an investment for n periods at the rate $i_n$

# EXAMPLE

| Period | Spot | Forward |
|---|---|---|
| 1 | 10% | |
| 2 | 10.50% | 11% |
| 3 | 10.80% | 11.40% |
| 4 | 11% | 11.60% |
| 5 | 11.10% | 11.50% |

$$f_0(1,2) = 1{,}105^2/1{,}1 = 11\%$$

$$f_0(1,2) = 1{,}111^5/1{,}11^4 = 11{,}5\%$$

# EXPECTATIONS THEORY

- Expectation theory holds that the forward rate is equal to the expectation of the spot rate that will prevail in the future.
- If $i_{1,1}$ spot rate on one-year bond at the beginning of period 1, $i_{1,2}$ spot rate on one-year bond at the beginning of, $i_{2,1}$ the spot rate on two-year bonds known at the beginning of period 1, $f_2$ forward $(1 + f_2) = \frac{(1+i_{2,1})^2}{1+i_{1,1}}$.
- According to the theory of expectations $f_2 = E\left(i_{1,2}\right)$ where $E$ is the expectation. If true, we can start from the term structure observed today to derive all forward rates and estimate market expectations regarding the future evolution of one-year rates

# EXPECTATIONS THEORY

- The expectation theory assumes that investors are risk-neutral and therefore prefer bonds with higher expected returns
- Excess demand for bonds with higher expected returns raises their price and brings the expected return back in line with that of other bonds
- One implication of the theory is that long-term and short-term bonds give the same expected return, and indeed that the expected return from any bond investment strategy is the same
- For example, the return obtained by investing in a two-year bond or in a series of one-year bonds would make the two strategies completely indifferent because:

$$(1 + i_{2,1})^2 = (1 + i_{1,1})(1 + Ei_{1,2})$$

# EXPECTATIONS THEORY

- Applying log to both sides:

$$2\ln\left(1+i_{2,1}\right)=\ln\left(1+i_{1,1}\right)+\ln\left(1+Ei_{1,2}\right)$$

i.e. approximately, the long-term rate is the average of the current short-term rate and the expected rate for the future

$$i_{2,1}=\frac{i_{1,1}+Ei_{1,2}}{2}$$

- Empirical tests regarding the validity of the theory of expectations show the existence of numerous deviations

# SEGMENTATION THEORY

- The segmentation (or habitat) hypothesis: investors have a preference for investing in certain maturities, for example they prefer short-term bonds (this hypothesis is known as the liquidity preference theory), and therefore need a premium to hold long-term bonds:

$$(1 + i_{2,1})^2 > (1 + i_{1,1})\,(1 + E i_{1,2})$$

$$i_{2,1} > \frac{i_{1,1} + E i_{1,2}}{2}$$

i.e. the term structure should be positively sloped

# SEGMENTATION THEORY

- Conversely, if investors have a preference for investing in long-dated bonds, such as 2-year bonds, two-year bond prices will be higher and their rates lower, therefore:

$$(1 + i_{2,1})^2 < (1 + i_{1,1})\,(1 + E i_{1,2})$$

or

$$i_{2,1} < \frac{i_{1,1} + E i_{1,2}}{2}$$

and the term structure will be sloping downwards to persuade investors, who would prefer two-year bonds on a like-for-like basis, to hold bonds with shorter maturities

# SEGMENTATION THEORY AND PREMIA

- In the case of segmentation, the interpretation of the relationship between rates over different time maturities can be made by considering premiums, i.e. returns demanded in excess of the risk-free rate that provide compensation for investor preferences
- We have seen how, based on the preference for short-term bonds, it can be found on the market that two-year rates are higher than the average current rate and those expected for the future $i_{2,1} > \frac{i_{1,1}+Ei_{1,2}}{2}$
- We interpret the spread between two-year rates on the one hand and current and future short-term rates on the other hand, as the result of a positive liquidity premium that must compensate for long-term bond holders:

$$(1+i_{2,1})^2 = (1+i_{1,1})\,(1+Ei_{1,2}+LP_2)$$

# SEGMENTATION THEORY AND PREMIA

- In this case, the forward rate is no longer equal to the expected rate but also includes a premium.

- We cannot therefore measure forward rates from the curve observed today and infer the expected short-term rates for the future

# SEGMENTATION THEORY AND PREMIA

- In the case of preference for longer maturities:

$$(1 + i_{2,1})^2 = (1 + i_{1,1})\,(1 + Ei_{1,2} + LP_2)$$

but in this case the premium is negative since investors are willing to give up a part of the yield in order to hold long-term securities

- If liquidity premiums are constant for the various maturities over time, it is easier to measure these premiums from the differential between the historical average of forwards and the historical average of sports rates. If liquidity premiums are variable over time, empirical measurement is much more difficult

# EXAMPLE

- Block 1: the theory of expectations applies
- Block 2: expectations about future rates are the same, but investors prefer to buy 3-year bonds, so their price rises and the three-year rate falls. The liquidity premium is -1.52%
- Block 3: investors prefer short maturities for which the price of long-term securities is lower and their spot rate is higher. The premium is equal to 1.54%, to be interpreted as additional compensation for holding securities whose maturity does not coincide with the preferred habitat

| Spot | Forward | Expectations | Premium |
|---|---|---|---|
| 1% | 1% | | |
| 2% | 3.01% | 3.01% | |
| 3% | 5.03% | 5.03% | |
| | | | |
| 1% | 1% | | |
| 2% | 3.01% | 3.01% | 0% |
| 3% | 5.03% | 6.03% | -1% |
| | | | |
| 1% | 1% | | |
| 2% | 3.01% | 3.01% | 0% |
| 3% | 5.03% | 4.03% | 1% |